

hw 0330

11.1

①

factor: type of fertilizer
levels: 3 types of fertilizers

②

factor: daily dose of vitamin C
levels: 500 mg, 500 mg, 1000 mg

③

not normal. The data are proportions based on counts out of 100 items, so the values are bounded between 0 and 1, usually binomial rather than normally distributed

11.2

④

a.

ANOVA

	df	SS	MS	F
treatment	2	655.52	327.76	4.11
error	8	638.3	79.79	
total	10	1293.82		

b.

$$F_{0.05, 2, 8} = 4.46$$

$$F = 4.11 < 4.46$$

Fail reject H_0 , no sufficient evidence

⑤

a.

	df	SS	MS	F
treatment	2	170.45	85.225	5.704
error	9	134.469	14.941	
total	11	304.919		

p-value = 0.0251 < 0.05
significantly different

b. $\bar{X}_A = 60.5, \bar{X}_B = 54.667$
 $60.5 - 54.667 = 5.833$
 5.833 ± 9.594
 $= (-3.761, 15.427)$

c. $(56.959, 64.043)$

d. No.
sample size is too small

(4)

a.

	df	SS	MS	F
treatment	3	4.996	1.599	8.43
error	22	4.172	0.19	
total	25	8.968		

b. $F_{0.05, 3, 22} = 3.05$

$F_{0.01, 3, 22} = 8.42$

$8.43 > 3.05$

$8.43 > 8.42$

Reject H_0

c. $(-0.84, 0.04)$

d. $(-0.36, 0.46)$

e. light tuna in water vs. white tuna
light tuna in oil vs. white tuna